

Evaluation Zero-shot generalization of validation strategies and transferability of EEG attention-state decoding : a comparative analysis across increasingly stringent validation strategies on open datasets

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Abstract

Despite growing interest, the development of robust EEG classification models remains a challenge. EEG signals are inherently noisy,

Electroencephalography (EEG) decoding models must generalize across people and experimental paradigms, yet non-stationary signals and the uncertain equivalence of labels such as “attention” make this difficult. Evidence on whether nominally aligned attention-related datasets provide useful zero-shot transfer across tasks and datasets remains limited. Here, we evaluated zero-shot generalization of an operationally defined binary contrast between high- and low-attention states in two technically compatible public EEG datasets: a five-participant mental attention-state dataset recorded during a train-simulator task and SAM-40, a 40-participant dataset spanning Stroop, arithmetic, and mirror-image tasks. The datasets were harmonized through a shared 12-channel input space, a fixed preprocessing pipeline, and a common label mapping. Drowsiness was excluded to avoid conflating the contrast with differences in arousal, visual input, and ocular activity. Five classifiers spanning linear, tree-based, and exhibit high inter-subject and inter-session variability.

Furthermore, EEG-based models often fail to generalize across different recording sessions, subjects, or task contexts, which limits their real-world applicability. One of the key contributors to this limitation is the lack of rigorous validation procedures and over-reliance on subject-dependent or intra-session evaluations that tend to overestimate model performance. In this work, two publicly available datasets recorded with technically compatible Emotiv headsets but distinct cognitive paradigms were aligned through a unified binary attention labeling scheme, a shared subset of 12 electrodes, and a standardized preprocessing pipeline including bandpass filtering, independent component analysis, short-time Fourier transform, spectral band decomposition, and overlapping window segmentation. Model performance was assessed under deep-learning approaches were assessed using group-disjoint within-dataset, cross-subject, cross-session, cross-task, and cross-dataset validation settings. Within-dataset evaluation yielded the strongest results, with accuracy and F1-score reaching 0.86 for Dataset A and 0.81 and 0.88, respectively, for Dataset B. Performance declined under more stringent transfer settings: cross-subject and cross-session validation showed moderate robustness, , without target-specific adaptation. Mean balanced accuracy across decoders was nearly unchanged when generalizing to unseen participants, differing from the within-dataset baseline by at most two percentage points. Transfer across heterogeneous domains was consistently costlier: cross-task transfer was only partially successful when tasks shared cognitive demands, fell on average 6.4 percentage points below the corresponding target-specific baselines and cross-dataset transfer remained the most difficult scenario, with one transfer direction approaching chance level. These findings indicate that current EEG classifiers capture attention-related features that are strongly shaped by experimental context rather than invariant neural signatures 13.7 percentage points below them, with a decline in every directed transfer comparison examined. Absolute accuracy also varied materially across decoders. Thus, shared channels, technically compatible acquisition, and nominally aligned labels were insufficient to preserve target-specific performance under heterogeneous transfer. These leakage-resistant, directed pre-adaptation benchmarks support empirical source selection and provide matched references for evaluating whether transfer learning, domain adaptation, or personalization yields meaningful gains. The findings concern protocol-defined contrasts rather than a unitary neural marker of attention. The study highlights the importance of rigorous validation protocols and underscores the need for future methods that improve cross-paradigm generalization in EEG attention decoding.

Introduction

Electroencephalography (EEG) has become a key modality in the development of is a technique for recording brain activity using multichannel scalp electrodes that measure voltage signals reflecting underlying cognitive processes. It is widely used in neuroscience and brain-computer interface (BCI) systems and cognitive monitoring tools, due to research because of its non-invasive nature, high temporal resolution, and ability to capture brain dynamics associated with cognitive states such as attention, workload, and emotional arousal. In particular, EEG-based attention recognition has gained increasing importance in applications ranging from neurofeedback and education to safety-critical systems such as drowsiness detection in transportation and adaptive human-machine interfaces.

Despite growing interest, the development of robust EEG classification models remains challenging. EEG portability, and relatively low cost [1]. At the same time, EEG signals are inherently noisy, non-stationary, and exhibit high inter-subject and inter-session variability. Furthermore, EEG-based models often fail to generalize across different recording sessions, subjects, or task contexts, which limits their real-world applicability. One of the key contributors to this limitation is the lack of rigorous validation procedures and over-reliance on subject-dependent or intra-session evaluations that tend to overestimate model performance.

In response to these issues, recent literature has emphasized the importance of evaluating EEG models under more realistic scenarios, including cross-subject, cross-session, and cross-task validation schemes. However, less attention has been paid to the problem of cross-paradigm transferability of neural markers, that is, whether EEG features learned in one task paradigm (e.g., passive observation with predefined attentional conditions) can generalize to another (e.g., active cognitive testing tasks). This question is particularly relevant for the development of generalizable attention-detection models.

Another major challenge is that many comparative EEG studies suffer from technical mismatches between datasets, such as differences in hardware, number of channels, electrode placement, or sampling rates, which introduce distributional shifts and confound performance assessments. For example, prior domain adaptation studies have shown poor transfer performance due to such technical discrepancies rather than inherent limitations of neural representation.

To address these gaps, this study proposes a comparative analysis of EEG-based attention classification across two open-access datasets recorded with the same EEG device (Emotiv headset), but designed under distinct cognitive paradigms. One dataset uses explicit attention labels in a continuous, passive-control setting, while the other captures attention-related processes through well-established cognitive tasks, such as the Stroop test and mental arithmetic.

By aligning these datasets conceptually and technically, we aim to examine the transferability of neural attention markers across paradigms, and systematically evaluate how different validation strategies impact model generalization.

Related work

The development of EEG-based classification models that generalize [2, 3]. In addition, these challenges are compounded by the scarcity of large, diverse, and well-annotated open EEG datasets [4]. Consequently, developing a robust decoding model with strong generalization capabilities across subjects, sessions, and tasks is an active area of research in brain-computer interface (BCI) and cognitive monitoring fields. A growing body of work has explored how different validation strategies affect the robustness and transferability of EEG models. This section reviews relevant studies on cross-validation, domain adaptation, and cross-task EEG analysis, with a specific focus on attention and cognitive state recognition. remains an open and central challenge [2, 5]. One promising approach to address these problems is transfer learning (TL). TL reuses representations already learned on one or more source problems to reduce the amount of labeled data and training effort a related target problem would otherwise require [6]. Its success depends not only on modeling and learning strategies but also on the degree of domain and label compatibility between source and target datasets.

For cognitive constructs such as attention, this compatibility is especially consequential and difficult to guarantee because attention remains an ambiguous concept and is not a unitary construct with a single universally accepted operational definition [4, 7]. Rather, attention is an umbrella term encompassing a set of partially independent processes and functions, such as selecting relevant information, maintaining focus over time, shifting attention between tasks, and allocating cognitive resources. These processes may operate on both external sensory input and internally represented information, including information maintained in working memory [7–9]. The relative contributions of these processes may vary with task demands, sensory modalities, task goals, experimental instructions, assigned conditions, self-reported assessments, and other aspects of experimental design and label definition. Consequently, experimental paradigms described as measuring “attention” may probe non-equivalent configurations of cognitive processes. As a result, different datasets may represent different attention-related cognitive states while assigning them the same nominal labels. In this study, we therefore treat “high-attention” and “low-attention” as operationally defined protocol conditions rather than as an independently validated attention construct, and ask whether decoders discriminate these conditions rather than whether they decode attention itself.

One of the most prominent lines of research involves domain adaptation techniques to reduce the discrepancy between training and testing distributions. For instance, [?] conducted a comprehensive evaluation of feature-based domain adaptation methods, including MIDA, TCA, GFK, and others, on two widely used emotion-related EEG datasets: DEAP and SEED. Their experiments included leave-one-subject-out and cross-dataset validation, where models were trained on one dataset and tested on the other. The results revealed limited improvements, which the authors attributed to substantial differences in recording hardware and data distributions between the datasets. This finding underscores the importance of technical consistency in cross-dataset evaluations, a factor addressed in the present study, where both datasets were collected using Emotiv EEG headsets with comparable specifications.

[10] proposed a hybrid feature extraction approach for classifying mental attention states under cross-session and Previous EEG transfer-learning research has been predominantly concerned with cross-subject scenarios. Their method combines spectral features obtained via short-time Fourier transform (STFT) with measures of functional and effective brain connectivity and employs a two-stage feature selection strategy to enhance model generalization. Experiments on two publicly available datasets demonstrated high classification accuracy (up to 94.01% in cross-subject evaluation), confirming the potential of integrating spectral and network-based features to improve the robustness of attention decoding. However, this work focuses on and cross-session generalization, whereas cross-task, cross-dataset, and cross-subject scenarios and does not address the transferability of models across different tasks, datasets, or paradigms. The present study is aimed at filling this gap by examining the extent to which cross-paradigm transfer have received comparatively less systematic attention [2, 11, 12]. Other recent work has evaluated cross-subject robustness of deep decoders on multiple motor-imagery benchmark datasets while keeping the task paradigm fixed [13]. Consequently, systematic empirical evidence for source–target suitability across heterogeneous attention-related neural markers retain their discriminative power when transferred across cognitive paradigms.

Several studies have investigated the cross-task generalization of EEG features. In a notable example, [?] analyzed the cross-task consistency of EEG-based mental workload indicators using a custom dataset involving two distinct paradigms: the N-back task and the Multi-Attribute Task Battery (MATB). The authors compared the performance of classifiers trained and tested within the same task versus across tasks. Their findings suggested that while task-specific models performed better, certain EEG features, especially in the power spectral domain, exhibited moderate stability across task contexts.

Task generalization has also been studied in attention recognition. For example, Nguyen et al. [?] proposed a Kalman filtering-based model and evaluated its robustness across multiple datasets. Their results demonstrated the feasibility of cross-dataset attention classification, although model performance degraded when shifting between tasks with differing structures.

Beyond feature-based alignment, a growing literature tackles domain shift in EEG with adversarial and deep domain-adaptation frameworks (e.g., domain-adversarial training with gradient reversal and adversarial discriminative adaptation), as well as personalization and few-shot calibration strategies that fine-tune a shared model to each new subject. Representation-learning approaches based on transformer/patching architectures and self-supervised or masked pretraining have likewise reported improved within- and cross-subject performance on cognitive EEG datasets. These methods are complementary to the present work: rather than proposing a new adaptation algorithm, we quantify how far conventional spectral features and a standard classifier transfer under increasingly stringent validation, thereby establishing a baseline against which such specialized methods can be compared.

While these studies highlight the potential for EEG-based cross-task transfer, they often suffer from two key limitations. First, many rely on custom or proprietary datasets, which reduces reproducibility and hinders broader benchmarking. Second, differences in data acquisition hardware often confound comparisons and make it difficult to isolate the true effect of task variability EEG datasets and tasks remains limited. Source–target suitability is often motivated conceptually or inferred from nominally aligned labels. This leaves unclear which source–target combinations provide useful zero-shot transfer, how much

target discriminability is retained relative to target-specific baselines, and whether transferability is direction-dependent. Establishing such pre-adaptation baselines is necessary to justify source selection and to interpret gains or negative transfer from subsequent transfer-learning and domain-adaptation methods [14].

In contrast, our study contributes to this line of work by focusing on attention-related EEG datasets that are both publicly available and technically compatible. We specifically investigate whether attention-related neural patterns, learned from data collected in one paradigm (passive observation with explicit labeling), can generalize to a different paradigm (active cognitive testing), without the confounding effects of sensor mismatch. To the best of our knowledge, comparative evaluations of cross-paradigm transferability that use open EEG datasets recorded with the same device family remain scarce, and the present analysis is intended as a step toward filling this gap rather than an exhaustive benchmark. Here, we address this gap by systematically evaluating zero-shot generalization across two technically compatible public EEG datasets spanning multiple tasks. Under a common preprocessing pipeline, we place within-dataset baseline performance, cross-subject generalization, and zero-shot transfer across tasks and datasets on a single ladder of increasingly stringent validation strategies. We characterize how performance behaves along this ladder, comparing directed source–target pairs against target-specific baselines to establish whether transfer is direction-dependent, and we test whether the resulting pattern is reproduced across five decoders ranging from linear and tree-based baselines to deep neural networks. Together, these analyses yield reproducible pre-adaptation benchmarks for evaluating candidate sources and contextualizing both gains and negative transfer produced by subsequent transfer-learning and domain-adaptation methods.

Open-access datasets

EEG data for mental attention state detection (Dataset A)

The dataset described in [15] comprises 34 EEG recordings collected from five participants 34 EEG recordings collected from five participants over the course of seven daysseven days, with the fifth participant completing only six sessions six sessions. During each session, participants operated a virtual train using the *Microsoft Train Simulator*. Each recording lasted approximately 35 to 55 minutes. All experiments in this study were performed with healthy volunteer participants chosen among students35 to 55 minutes. The participant cohort consisted of healthy university student volunteers.

To ensure consistency, all sessions were conducted on During every session, participants followed the same predefined route , designed to require minimal interaction. This setup constitutes a low-intensity control task, minimizing variability due to task demands and aligning with naturalistic stimulus-based protocols as described in [10]in the virtual train simulator. The route required little active control and kept task demands comparable across recordings. Each session was divided into three sequential phases, each comprised three sequential phases designed to reflect a different target mental state:

focused, unfocused, and drowsy. The participants simulated these mental states based on standardized instructions provided by distinct target mental states: *focused, unfocused, and drowsy*. Participants received standardized instructions from the experiment supervisor for each phase:

- **Focused (0–10 min):** Participants actively monitored and mentally engaged The focused phase required active monitoring of and mental engagement with the simulator, maintaining including sustained attention to the virtual controls and screen events.
- **Unfocused (10–20 min):** Participants During the unfocused phase, participants disengaged from the task, ceasing interactionand intentionally avoiding attentional focus, while being ceased interaction, and were instructed to keep their eyes open and remain awake.
- **Drowsy (20–30 min):** Participants During the drowsy phase, participants were allowed to relax, close their eyes, and enter a drowsy or lightly dozing state, simulating reduced cognitive arousal.

EEG signals were recorded using a modified Emotiv headsetmodified Emotiv headset, equipped with wet electrodes and supporting wireless Bluetooth transmission. The device recorded data at a sampling rate of 128 Hz sampling rate of 128 Hz and with a voltage resolution of 0.51 μ Vvoltage resolution of 0.51 μ V. The dataset includes 14 EEG channels14 EEG channels, labeled according to the 10–20 international system: AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4.

SAM-40: dataset of 40-subject EEG recordings (Dataset B)

The SAM-40 dataset, described in [16], contains EEG recordings from 40 healthy participants (14 females and 26 males) with a mean age of 21.5 years. The purpose of data collection was It was originally collected to investigate short-term stress responses induced by the performance of various cognitive tasks, which correspond to a . Its protocol corresponded to the continuous cognitive engagement paradigm as described in [10] . The experimental protocol and included four conditions:

- **Resting state**— EEG recording during relaxation : Participants relaxed while listening to music during EEG recording.
- **Stroop Color-Word Test**— participants were required to identify : Participants identified the color of the ink in which a word was printed , while ignoring the word’s semantic meaning. The task included two types of stimuli: *congruent* (the color name matched the ink color) and *incongruent* (the color name and ink color differed). Each execution of the task task run included 11 stimuli.

- **Arithmetic problem solving**— participants : Participants mentally solved a mathematical problem and determined whether the presented answer was correct, responding with a “thumbs up” or “thumbs down” gesture. Each execution of the task task run included 6 arithmetic stimuli.
- **Mirror Image Recognition**— participants : Participants were shown images and asked to determine whether they were mirror-symmetric or asymmetric, responding with the same gestures. Each execution of the task task run included 8 images.

Each task lasted was performed for 25 seconds, with 5-second rest intervals and a 10-second instruction period preceding . Between consecutive cognitive tasks, participants rested for 5 seconds and then received 10 seconds of instructions for the next task. After the completion of each trial, participants rated the level of stress experienced for stress experienced during each task on a 10-point scale. For each participant, three trials were recorded using different sets of stimuli, where 1 indicated minimal stress and 10 indicated maximal stress. Three trials, each using a different set of stimuli, were recorded for every participant.

EEG signals were recorded using an Emotiv Epoc Flex headset with 32 wet electrodes, labeled according to the international 10–20 system. Data were acquired at a sampling rate of 128 Hz and a voltage resolution of 0.51 μV . The 32 channels included in the dataset are: Cz, Fz, Fp1, F7, F3, FC1, C3, FC5, FT9, T7, CP5, CP1, P3, P7, PO9, O1, Pz, Oz, O2, PO10, P8, P4, CP2, CP6, T8, FT10, FC6, C4, FC2, F4, F8, and Fp2.

Materials and methodsMethodology

The methodological design of this study was developed to evaluate the generalizability and transferability of EEG-based attention decoding across different recording conditions, participants, and task paradigms. To this end, we adopted two publicly available datasets that differ in experimental protocols but share compatible technical specifications, which allows for thereby enabling a controlled comparative analysis. The datasets were harmonized through a unified labeling scheme that distinguishes between low-attention and high-attention states, enabling consistent evaluation across scenarios. The datasets were harmonized through a binary operational labeling scheme that distinguishes high-attention states, defined by protocol-imposed task demand, from low-attention states of rest or task disengagement, enabling consistent evaluation across scenarios. This section outlines the rationale for dataset alignment, the strategies employed for cross-validation, and the experimental settings designed to assess within-dataset performance as well as cross-subject, cross-task, and cross-dataset generalization.

All experiments were executed through a version-controlled computational workflow. For every run, the workflow stores the resolved configuration of data preparation, validation, and model settings together with the run outputs needed to reconstruct the reported evaluation metrics. These records provide run-level provenance

and allow each reported metric to be traced to the experiment that produced it. The complete implementation, experiment configurations, and run-level artifacts are available at <https://github.com/sovetskiysn/eeg-validation-strategies>, released under the MIT license and archived at <https://doi.org/10.5281/zenodo.22255472>.

Dataset alignment and integration rationale

Effective evaluation of cross-task and cross-dataset generalization relies on the availability of datasets that are compatible both technically and conceptually. In this study, we integrate used two publicly available EEG datasets that differ in their original purpose and labeling schemes, yet share critical properties enabling a unified analysis of attentional states retain sufficient technical and conceptual compatibility for a meaningful comparison.

A major strength of this dataset combination lies in their acquisition setup is the compatible acquisition setup of the two datasets. Both datasets were recorded using consumer-grade Emotiv EEG headsets with identical electrode configurations, Emotiv EEG systems with the same sampling rate (128 Hz), and voltage resolution. This uniformity eliminates a key source of variability, namely hardware-induced distribution shifts, that frequently, although their original electrode montages differed (14 versus 32 channels). For cross-dataset analyses, only the 12 channels common to both datasets were retained. Differences in EEG recording systems and electrode configurations can hinder cross-dataset transferability as shown by [?]. Consequently, performance differences observed in our experiments can be attributed primarily to task and participant factors, rather than to mismatched recording systems. [17]. Restricting the analysis to the shared channels substantially reduces this configuration-related mismatch, although it does not eliminate all acquisition-related distribution shifts. Within the limited set of publicly available datasets suitable for this comparison, this pair offered a comparatively compatible basis for the present comparative evaluation.

Although the first The conceptual rationale for combining the datasets rests on the cognitive conditions represented in their respective protocols. The mental attention-state dataset contains explicit attention state labels (*Focused*, *Unfocused*, and *Drowsy*) collected during a low-interaction simulation task, and the second whereas SAM-40 was originally designed for stress assessment, both capture cognitive conditions strongly tied to attentional processes. The second . Both include conditions involving attentional processes, although their original constructs differ. The SAM-40 dataset includes tasks widely recognized in cognitive neuroscience for eliciting high levels of as requiring attentional engagement:

- **Stroop Color-WordTest** — : probes both *selective* and *sustained attention*, requiring participants to suppress automatic reading responses [?, 18, 19] and continually focus on the ink color throughout the task [18–20].

- **Mental arithmetic task**— : demands *sustained internal attention* and working memory for continuous manipulation of numerical information [?, 21] [21, 22].

- **Mirror image recognition task**Image Recognition— : engages visuospatial processing and requires both *selective* and *sustained attention* to discriminate between symmetric and asymmetric patterns [?, 23, 24], relying on mental rotation processes that demand continuous cognitive effort [23–25].

These two datasets correspond to the simulator-based and task-based datasets, respectively, to the naturalistic stimulus-based protocols and continuous cognitive engagement paradigms described in [10]. In this unified labeling scheme, conditions such as *Rest* or *Rest* and *Unfocused* are treated as low-attention states, whereas *Focused*, *Stroop*, *Arithmetic*, and *Mirror recognition* are treated as high-attention states. The *Drowsy* condition of Dataset A denotes a state of reduced cognitive arousal and disengagement from the task; it was therefore mapped to the low-attention class together with *Unfocused* and *Rest*, rather than being discarded. The *Drowsy* phase is excluded from this mapping and from all subsequent preprocessing and analyses because permitted eye closure and drowsiness may confound the comparison with differences in arousal, visual input, and ocular activity. This mapping is summarized in Table ?? 1. Throughout this paper, “high-attention” and “low-attention” states denote these operationally defined protocol conditions rather than an independently validated attention construct.

Experimental design

The experimental design was structured to capture different dimensions of model generalization , ranging from subject- and session-level variability to cross-task and cross-paradigm transferability. Instead of relying on a single evaluation scheme, we combined five complementary validation strategies, each addressing a distinct source of variability.

Validation strategies

Baseline (within-dataset evaluation). Within-dataset performance was estimated using stratified K-fold cross-validation applied separately to each dataset. This setting provides an upper bound of model performance when training and testing distributions are identical, serving as a reference for subsequent generalization analyses. Unless otherwise stated, K was set to 5. Because the continuous recordings were segmented into densely overlapping windows (see Data preparation), windows derived from the same recording are temporally autocorrelated; we discuss the resulting risk of optimistic within-dataset estimates in the Limitations subsection. We evaluated the decoders under four validation protocols, each addressing a distinct generalization question. To preserve the independence of the training and test data, all windows originating from the same recording unit were assigned to the same partition. This prevents overlapping windows from being divided between training and test sets, reducing the risk of overly optimistic performance estimates due to temporal autocorrelation between

neighboring EEG windows [10]. The within-domain baseline estimates performance when training and test data share the same dataset and condition composition but originate from different recording units. The remaining protocols examine whether discriminative performance is retained when training and testing are separated by participant, task composition, or dataset. Each protocol was used only where the structure of the relevant dataset supported its interpretation. For example, cross-task validation was possible only for SAM-40 because it included multiple task compositions. Model fitting was confined to the training data, and the resulting decoder was evaluated on held-out data without target-specific adaptation. Any class imbalance was addressed through inverse-frequency class weighting calculated independently within each training fold, while test data retained their observed class distribution.

Baseline evaluation. Reference performance was estimated using five-fold stratified group cross-validation, separately for the mental attention-state dataset, the SAM-40 dataset, and each task-specific composition of SAM-40. The grouping variable was the physical recording unit: a selected session in the mental attention-state dataset or a run-level recording unit in SAM-40. Each SAM-40 unit comprised all task recordings included in the analysis for that run. Training and test folds thus shared the dataset and condition composition and could contain data from the same participants, but never from the same recording unit. Keeping each unit intact prevented temporally adjacent or partially overlapping windows from the same acquisition from being assigned to opposite folds, thereby avoiding window-level temporal leakage. These estimates serve as within-domain comparators for the transfer protocols.

Cross-subject validation. Generalization across individuals was examined using a leave-one-subject-out (LOSO) protocol. In each iteration, data from one participant were held out for testing, while the model was trained on all remaining participants. This evaluation reflects the model’s ability to adapt to previously unseen subjects. In each leave-one-subject-out fold, the decoder was trained on all selected data from the remaining participants and tested on all selected data from the held-out participant. The resulting performance quantifies zero-shot generalization to an unseen participant.

Cross-session validation. To test robustness against intra-individual variability (e.g., electrode placement shifts, fatigue, or day-to-day differences), a leave-one-session-out scheme was adopted for the multi-session dataset. Here, the model was trained on all but one recording session of a participant and tested on the held-out session.

Cross-task validation. Cross-task generalization was assessed within Dataset B, which includes multiple cognitive tasks (Stroop, Arithmetic, and Mirror recognition). The model was trained exclusively on one task and evaluated on another (e.g., Stroop → Arithmetic). This protocol probes whether neural markers of attention learned in one cognitive paradigm retain discriminative power in a different paradigm. Similar cross-task transfer analyses have been conducted in prior studies on EEG workload and attention [?]. In the present cross-task experiments the source and target tasks were drawn from the same pool of Dataset B participants; consequently, task transfer and subject identity are not fully disentangled, a confound we revisit in the Limitations subsection. In SAM-40, each

task composition represented the same binary contrast of low-attention rest versus high-attention task conditions, rather than classification of task identity. Cross-task transfer therefore asked whether a decoder fitted to this contrast under one task composition retained discriminative performance under another.

The cross-task folds were also subject-disjoint. For each fold, the source composition from all but one participant formed the training set, whereas the target composition from the held-out participant formed the test set. The protocol thus evaluates the combined shift to an unseen participant and a different task composition, rather than within-participant transfer between tasks. This distinction prevents participant-specific EEG patterns from contributing to apparent cross-task generalization [26, 27].

Cross-dataset validation. Finally, cross-dataset transferability was investigated by training models on one entire dataset and testing them on the other (Dataset A $\rightarrow$ Dataset B, and vice versa). Additionally, subset experiments were performed (e.g., training on Dataset A and testing only on Stroop trials of Dataset B). This protocol can also be interpreted as a form of *cross-paradigm* evaluation in the taxonomy proposed by [10]. The approach follows established cross-dataset validation protocols used in EEG research [?, ?]. To evaluate transfer between datasets, models were trained on all available data from a source composition and tested on a composition from the other dataset, in both transfer directions and without target-specific adaptation. SAM-40 was evaluated both as a complete composition and as task-specific compositions. Unlike the preceding protocols, cross-dataset validation changes several factors simultaneously, including the participant population, experimental paradigm, recording setup, and condition definitions. It therefore measures transfer under their compound distribution shift rather than attributing a performance change to any single factor [17]. The common sampling rate, harmonized binary labels, and shared 12-channel montage establish a compatible input and prediction space, but do not remove the substantive differences between the two datasets.

For performance evaluation, we report four widely used classification metrics: *Accuracy*, *Precision*, *Recall*, and *F1-score*. To ensure fair assessment in the presence of class imbalance, particularly in Dataset B, *Precision*, *Recall*

Data preparation

The same preparation pipeline was applied to both datasets and across all validation protocols. Its filtering, ICA, channel alignment and windowing parameters were fixed in advance and were not tuned separately for a dataset, task composition, validation protocol or decoder. The purpose was to place recordings in a common input space while limiting preprocessing itself as an additional source of variation. Differences between validation results are therefore interpreted against a fixed preparation procedure. The pipeline was instantiated as a fixed set of dataset preparations, that is, the analytical compositions used in every validation protocol and listed in Table 2. The remainder of this subsection describes the selection, cleaning, alignment and segmentation steps that produced them.

Recording and label selection. Recordings were selected before signal processing. Following the source study, the first two recording sessions for each participant in the mental attention-state dataset were intended for habituation to the simulator and were not included in the analyses [15]. The *Drowsy* condition was excluded from the binary classification analysis because participants were permitted to close their eyes and enter a drowsy state. This condition differs from the *Focused* and *Unfocused* conditions not only in protocol-defined task demand, but also in arousal, visual input and ocular activity. Including it in the low-attention class would therefore confound the intended contrast. The drowsy segment was removed before filtering and ICA, so no drowsy epochs were generated or presented to a classifier. In contrast, the complete collection of SAM-40 recordings was retained for analysis. SAM-40 provides the task recordings of each trial as separate files. Within each scenario, the task recordings selected from one trial were grouped into a single recording unit, whereas task recordings not selected by the scenario were not carried forward to later preparation steps. Accordingly, the binary mapping in Table 1 was applied to the retained recordings from the two datasets.

Filtering and ICA cleaning. Filtering was applied to each recording, and ICA was fitted within participant-specific acquisition groupings. For the mental attention-state dataset, each grouping corresponded to a selected session, whereas for SAM-40 it comprised the selected task recordings of a participant. To prevent preprocessing from becoming an additional source of variation, we used a deliberately restrained, conventional procedure: standard filtering followed by ICA-based artifact cleaning. Keeping preparation deliberately simple allows differences in performance to be interpreted primarily in relation to the recordings and validation protocols, rather than to a complex preparation pipeline. EEG channels were notch-filtered at 50 Hz, band-pass filtered from 1 to 45 Hz and rereferenced to the average reference. Artifact correction used independent component analysis with extended-Infomax optimization, retaining components that explained 99% of the variance. The resulting components were automatically classified from their spatial, spectral and temporal characteristics. Components identified as eye-blink or muscle artifacts with a probability of at least 0.80 were removed. The ICA random state was fixed by the experiment-wide configuration.

Channel alignment and window segmentation. After signal cleaning, each recording was restricted to the 12 channels shared by the two datasets: F3, F4, F7, F8, FC5, FC6, O1, O2, P7, P8, T7 and T8. Each retained labeled block was segmented into 5-s windows with 0.5-s overlap. Every retained window inherited the label of its source block and metadata identifying its dataset, participant, session, task, run, recording unit and temporal onset. The recording-unit identifier was used by the validation protocols to ensure that all overlapping windows from a unit remained on the same side of a split.

The fixed selection, cleaning, alignment and segmentation steps therefore yielded a deterministic analytical window set for each dataset composition. All model-specific input representations were derived from these same

prepared windows, rather than from separately prepared data. The retained recording units, window counts and observed class distributions of the resulting compositions are those reported in Table 2. Class weights were calculated independently within each training fold. All Dataset B configurations contain the same 40 participants and differ only in the task recordings retained alongside the resting-state recordings.

Models

We evaluated five classifiers rather than relying on a single decoder to test whether the conclusions are robust to a change in modeling assumptions. This makes the decoder an explicit factor of the design: a difference between validation protocols that is reproduced across models is more credibly attributed to the data and evaluation protocol than to the idiosyncrasies of one learner. The selected models span three families used in EEG decoding: linear, tree-based, and deep-learning architectures. The deep networks are included because they learn task-relevant representations directly from the channel-by-time EEG windows, reducing reliance on a manually specified feature set. All models were trained to predict whether a window belonged to a high-attention or a low-attention state in the sense of Table 1.

Hyperparameters were fixed *a priori* and held constant across datasets, task compositions, validation protocols and runs. No hyperparameter search was performed. Values followed library defaults or published reference recipes, except for the departures specified below. A fixed random seed governed fold construction, model initialization and network training. The resolved configuration of each run was stored with its outputs.

Traditional machine learning. Both classical classifiers used the same fixed, label-free feature representation. For each of the 12 channels, we computed log relative power in the 1–4 Hz, 4–8 Hz, 8–13 Hz and 13–45 Hz bands, together with sample entropy, mean, variance, skewness and kurtosis. Relative band power was normalized by the channel total before log transformation. The nine features from each channel formed one 108-dimensional feature vector per window.

Logistic regression provides a transparent linear baseline. It estimates a linear decision boundary from the feature vectors, allowing the contribution of individual features to be inspected. Its feature vectors were z-score standardized using the mean and standard deviation estimated from the training fold only.

XGBoost [28] represents tree-based ensemble learning, allowing nonlinear relations and feature interactions to be modeled within the same feature representation. It was also the decoder used in the originally submitted version of this study, preserving a direct point of comparison with those results. It received the same feature vectors without scaling. For both classical models, inverse-frequency class weights were calculated from the training fold only.

Deep learning. The remaining three classifiers operate directly on the prepared channel-by-time EEG windows,

rather than on the classical feature vectors. Each input window had a 12×640 channel-by-time shape and was standardized using statistics estimated from the training fold only.

ShallowFBCSPNet [29] is a shallow convolutional neural network designed to emulate the main operations of the filter-bank common spatial pattern (FBCSP) pipeline within a trainable model. It first applies temporal convolution and spatial filtering across channels, then uses a squaring nonlinearity, mean pooling and a logarithmic activation to form band-power-like features. Thus, its inductive bias reflects a conventional EEG decoding approach, while its filters are optimized jointly with the classifier.

EEGNet [30] is a compact convolutional neural network designed specifically for EEG. Its temporal convolutions learn filters over time, depthwise convolutions learn spatial filters across electrodes for each temporal feature map, and separable convolutions combine these learned features efficiently. This factorized design substantially limits the number of trainable parameters while retaining separate temporal and spatial feature learning.

EEG Conformer [31] combines a convolutional front end with a Transformer encoder. The convolutional module forms local temporal and spatial features and converts them into a sequence of embeddings. Self-attention in the Transformer then models global dependencies between these embeddings. It therefore complements the local receptive fields of convolution with a mechanism for relating EEG patterns over longer time spans. To limit the risk of overfitting in the smallest group-disjoint training folds, we reduced the capacity of the Braindecode default configuration before evaluation and then fixed it across all scenarios. Specifically, we reduced the number of temporal convolution filters, which also determines the embedding width, from 40 to 16, the number of Transformer encoder layers from 6 to 2, and the number of attention heads from 10 to 4. This retained the convolution-attention architecture while reducing its parameter count for the 12×640 input windows from 527,746 to 170,466.

All three networks were trained from scratch, with no pretrained weights, for 100 epochs with a batch size of 64. Their loss functions used inverse-frequency class weights calculated from the training fold only, and no resampling was applied.

Results and analysis

To characterize zero-shot generalization across the prespecified validation strategies, we evaluated the binary contrast between high-attention and low-attention states with five decoders. Table 3 reports decoder-specific performance across these strategies. It presents participant-level balanced accuracy, macro F1, MCC, ROC-AUC, and *F1-score* were computed using a weighted averaging scheme. per-class recall for logistic regression, the interpretable reference decoder. Corresponding tables for XGBoost, EEGNet, ShallowFBCSPNet, and EEGConformer are

provided in S1 Table, S2 Table, S3 Table and S4 Table. 397

Data preparation 398

As the two datasets were recorded with different electrode montages (14 vs. 32 channels), only the common subset of 12 electrodes was 399
retained for analysis: F3, F4, F7, F8, FC5, FC6, O1, O2, P7, P8, T7, 400

Generalization under increasingly stringent validation strategies 401

To examine generalization under increasingly stringent validation strategies, we compared directed cross-task and 402
T8. Restricting the analysis to this overlapping set ensured consistency in the spatial representation of EEG signals and eliminated 403
potential biases due to non-matching channel configurationsCross-dataset transfer with the baseline scenario for each target 404
composition. These comparisons place performance under progressively less similar training and test data in 405
relation to a target-specific reference. They therefore assess whether the binary contrast was retained without 406
target-specific adaptation as the participant, task composition, or dataset differed between training and test data. 407

Following channel selection, all EEG recordings were preprocessed using a standardized pipeline. Signals were bandpass-filtered in 408
the range of 0.5–45 Hz to suppress line noise. Independent Component Analysis (ICA) was subsequently applied to identify and remove 409
components associated with ocular artifacts. Components were flagged for removal based on their scalp topographies and time courses 410
characteristic of eye-blink and lateral-eye-movement activity (frontal, low-frequency, high-amplitude patterns); typically two such 411
components were excluded per recording. Given that only 12 channels were retained, this low-density configuration limits the stability of 412
ICA decomposition; we note this constraint in the Limitations subsection. Fig 1 summarizes the directed cross-task and 413
cross-dataset comparisons. Rows are grouped by source composition and decoder, and columns denote target 414
compositions. Each nonmissing matrix cell reports balanced accuracy calculated separately from the held-out 415
windows of each target participant and then averaged across target participants for the given decoder and 416
direction. Off-diagonal cells represent zero-shot transfers without target-specific adaptation, whereas the gray 417
diagonal reports the target-specific baseline estimated with five-fold stratified group validation that keeps 418
recording units disjoint between training and test folds. For each target represented in a summary column, we first 419
averaged participant-level estimates across the five decoders. We then averaged these decoder-level transfer and 420
baseline values across targets and calculated their difference. Negative values therefore indicate lower transfer 421
performance than the applicable target baselines, whereas dashes identify comparisons that are not applicable. 422

After preprocessing, Fig 2 compares baseline and leave-one-subject-out cross-subject validation for the mental 423
attention-state dataset and the EEG signals were transformed into the time–frequency domain using the Short-Time Fourier 424
Transform (STFT). Instead of analyzing the entire broadband spectrum, SAM-40 dataset. For each decoder and scenario, 425

balanced accuracy was calculated separately from the held-out windows of each target participant and then averaged across target participants, and the colored lines show these decoder-specific means. Black points show the mean of the resulting spectrograms were decomposed into five canonical frequency bands most relevant for cognitive load and attentional processes: *delta* (0.5–4 Hz), *theta* (4–8 Hz), *alpha* (8–13 Hz), *beta* (13–30 Hz), and *gamma* (30–45 Hz).

The continuous EEG recordings were segmented into overlapping windows of 1 second duration with a step size of 0.125 seconds. This windowing strategy approximates the requirements of real-time applications while maintaining sufficient temporal and spectral resolution for robust analysis five decoder-specific estimates. Their bars show paired 95% percentile bootstrap intervals from 10,000 target-participant resamples, with the same resample used for all decoders to preserve their pairing.

Classification model and training configuration

Classification was performed with a gradient-boosted decision-tree model (XGBoost), which is well suited to the resulting tabular, high-dimensional band-power feature representation and has been reported to perform robustly in EEG decoding tasks. Per-band, per-channel STFT power features were standardized prior to training. The same model family and feature representation were used across all validation scenarios so that differences in performance reflect the validation protocol rather than changes in the learner. The complete preprocessing and evaluation pipeline, including the exact model hyperparameters and data splits, is publicly available at <https://github.com/sovetskiysn/eeg-validation-strategies>.

The overall decay of performance as validation becomes more stringent is illustrated in Fig ??,

For cross-task transfer, the mean decreases from the corresponding target-specific baselines, averaged across all five decoders, were 9.6, 6.3, 6.6, and 3.1 percentage points for the Stroop, Arithmetic, and Mirror task compositions and the SAM-40 dataset, respectively. The mean decrease across these targets was 6.4 percentage points. For cross-dataset transfer, the full set of pairwise transfer outcomes within Dataset B is shown in Fig ?. decoder-averaged decreases for the Stroop, Arithmetic, and Mirror task compositions, the SAM-40 dataset, and the mental attention-state dataset were 13.1, 10.4, 13.1, 11.0, and 20.9 percentage points, respectively. The mean decrease was 13.7 percentage points. Thus, zero-shot transfer yielded lower performance in every eligible comparison, with a larger reduction when training and test data came from different datasets.

Results and discussion

All experimental results are summarized in Table ?. The dataset introduced in [15] is denoted as *Full A*, while the dataset described in [16] is denoted as *Full B*. This unified presentation facilitates direct comparison of validation schemes and transfer scenarios across both datasets. In the mental attention-state dataset, mean balanced accuracy was 65.5% in the baseline scenario and 65.4% in cross-subject validation, a decrease of 0.1 percentage points. In the SAM-40 dataset, it decreased from

69.1% to 67.1%, a difference of 2.0 percentage points.

The evaluation revealed a consistent degradation of performance as validation protocols became more stringent. Within-dataset validation (stratified K-fold) yielded the highest scores, with Dataset A reaching 0.86 in all metrics, and Dataset B performing comparably with 0.81 accuracy and 0.88 F1-score. Notably, when Dataset B tasks were considered separately, performance was lower for Stroop (0.78 F1), Arithmetic (0.72 F1), and Mirror (0.76 F1). However, when combined into a unified dataset, the model achieved superior overall performance, particularly in recall (0.97). This high recall, paired with a lower precision and F1-score, is consistent with a class imbalance in the harmonized Dataset B and a decision boundary biased toward the majority (high-attention) class; reporting of class distributions and uncertainty is addressed as a limitation below. This suggests that task aggregation amplifies the salience of attentional markers by capturing complementary neural mechanisms across paradigms, resulting in stronger discriminability than task-specific decoders.

With two exceptions, all decoder-specific estimates fell within the corresponding 95% bootstrap interval. In the SAM-40 baseline scenario, ShallowFBCSPNet reached 72.0%, which was slightly above the interval upper bound of 71.6%. In cross-subject and cross-session evaluations, accuracy dropped to 0.63–0.64 for Dataset A and 0.73–0.76 for Dataset B, reflecting the well-documented challenges of inter-individual variability and session non-stationarity in EEG data. Despite this degradation, Dataset B maintained relatively high F1-scores (up to 0.85), suggesting that task-induced attentional engagement provides more transferable features across individuals. These findings align with previous studies emphasizing the fragility of within-subject models when exposed to distributional shifts and highlight the potential of task-driven paradigms for mitigating such variability [?, 10] validation, EEGConformer reached 70.7%, which was above the upper bound of 69.7%.

Performance consistency across decoders

To assess whether performance patterns across validation strategies depended on decoder choice, Fig 3 compares all validation scenarios across the five decoders. Each colored point represents the balanced accuracy of one decoder, averaged across target participants in a given scenario. The black point and horizontal bar show the mean across the five decoders and its paired 95% percentile bootstrap confidence interval (CI), respectively, estimated from 10,000 target-participant resamples. The same resample was used for all decoders to preserve their pairing.

Cross-task transfers showed that generalization strongly depends on the cognitive similarity of tasks. Transfers between Stroop and Mirror achieved moderate success (up to 0.78 accuracy), suggesting that these paradigms engage partially overlapping attentional mechanisms. In contrast, transfers involving Arithmetic were consistently weaker (0.54–0.59 accuracy). This discrepancy likely arises from the substantial differences between the neurocognitive tasks: while Stroop and Mirror share elements of selective attention and visual discrimination, Arithmetic introduces a fundamentally distinct mode of processing. These results emphasize that the heterogeneity of task demands directly impacts the stability of attention-related neural markers and limits their transferability.

Cross-dataset transfers remained the most challenging scenario. Training on Dataset A and testing on Dataset B yielded moderate

accuracy (0.75), but the relatively low F1-score (0.64) indicates a bias toward the majority class, suggesting limited extraction of task-invariant attention features. Conversely, transfers in the opposite direction collapsed to chance-level performance (0.49–0.50 across metrics), underscoring the strong dataset-specific nature of EEG representations. These findings further demonstrate that while cross-subject, cross-session, and even certain cross-task evaluations suggest partial preservation of attention-related neural markers, such markers do not necessarily generalize across datasets . The near-random performance observed in Across the 27 scenarios, 42 of 135 decoder-specific estimates lay outside the corresponding 95% bootstrap interval. Logistic Regression lay outside the interval in seven scenarios, with four points below and three above the interval. XGBoost lay outside the interval in four scenarios, with one point below and three above the interval. All of these departures occurred in cross-dataset settings highlights the fragility of these features when confronted with paradigm-level variability. transfer settings.

The cross-dataset results reported here characterize the difficulty of the problem under a conventional pipeline; they do not represent an upper bound. Dedicated domain-adaptation methods, including correlation alignment (CORAL), transfer component analysis (TCA), domain-adversarial training with gradient reversal (DANN), adversarial discriminative domain adaptation (ADDA), and Riemannian alignment, as well as few-shot per-subject calibration, are explicitly designed to mitigate the distribution shift observed here and could plausibly raise cross-paradigm performance above chance. Because such adaptation and personalization baselines were not implemented in this study, the present cross-dataset numbers should be read as a baseline lower bound rather than evidence that the transfer problem is intractable. EEGNet lay outside the interval in 12 scenarios, always below it. ShallowFBCSPNet lay outside the interval in 10 scenarios, always above it. These were the two decoders with the most frequent departures from the scenario-average interval.

Taken together, these results illustrate that while EEG-based models can reliably capture attentional states within a single dataset, their performance does not extend robustly across subjects, sessions, tasks, or paradigms. The observed failures in transferability emphasize that current classifiers exploit features tied to specific experimental conditions rather than invariant neural signatures of attention. Addressing this limitation will require methodological advances capable of disentangling paradigm-specific effects from more generalizable neural markers EEGConformer lay outside the interval in nine scenarios, with two points below and seven above the interval. Overall, these patterns show that decoder choice materially influenced balanced accuracy across validation scenarios, including both within-dataset baselines and transfer settings.

Limitations and future work

Discussion

Several limitations qualify the present findings and motivate future work. First, the within-dataset baselines were computed over densely overlapping 1-second windows (0.125-second step); windows from the same recording are highly autocorrelated, so stratified K-fold that

splits such windows across folds can leak temporally adjacent samples and inflate within-dataset metrics. Group-aware splitting by trial, session, or subject, or temporal blocking, would yield more conservative and realistic estimates. Second, all reported numbers are point estimates: class distributions after label harmonization, fold-wise variability, confidence intervals, and statistical significance tests are not reported, which limits the strength of quantitative comparisons. Third, in the cross-task experiments the source and target tasks share the same Dataset B participants, so task transfer is not fully separated from subject identity; subject-disjoint cross-task splits are needed to isolate the task effect. Fourth, the low-density 12-channel montage constrains the stability of ICA-based artifact removal. Finally, the study does not include domain-adaptation or personalization baselines or additional diagnostic visualizations (e.g., confusion matrices, per-subject variability, or low-dimensional embeddings of the feature space). Addressing these points, releasing standardized leakage-free splits, and benchmarking adaptation methods are natural next steps that would strengthen the evidential value of this analysis. Across the data compositions and validation protocols examined here, zero-shot discriminative performance declined as the difference between the training and test data increased. Performance was comparatively stable when generalization was limited to an unseen participant within a dataset, but was lower when the training and test compositions differed by task and lower still when they came from different datasets. Thus, for these operationally defined high- versus low-attention contrasts, technical compatibility and a common binary label space were insufficient to preserve target-specific performance under increasingly heterogeneous zero-shot transfer conditions.

Conclusion

This work systematically evaluated EEG-based attention classification under multiple validation schemes, ranging from within-dataset baselines to the most challenging. The overall degradation pattern was observed across the five decoders, but the absolute level of balanced accuracy varied materially with decoder choice. This finding makes the decoder an important factor in comparative EEG evaluation rather than an interchangeable implementation detail. In particular, conclusions about a validation scenario should distinguish a pattern that is reproduced across model families from the performance of any single predeclared model recipe.

These results provide leakage-resistant, pre-adaptation baselines for assessing source–target suitability in the studied data compositions. They indicate that nominally aligned labels and partially harmonized acquisition specifications do not by themselves establish that a source dataset will support useful zero-shot transfer to a target dataset or task composition. Such baselines can therefore inform source selection, which should be based on empirical, directed source–target comparisons rather than on nominal label alignment alone. By relating each transfer result to the corresponding target-specific baseline, the analysis distinguishes preservation of the contrast under transfer from the discriminability of that contrast in the target data itself. The resulting references also make it possible to evaluate whether subsequent transfer-learning, domain-adaptation, or personalization methods

provide a meaningful improvement over matched zero-shot performance rather than merely achieving a favorable absolute score.

Several limitations delimit the interpretation of these findings. First, the binary labels represent protocol-defined contrasts and do not isolate attention from related differences in arousal, visual input, stress, or task demands. Second, selection of only two technically comparable datasets enabled a common input space for cross-dataset transfers. The results reveal a consistent pattern: while within-dataset models achieve high accuracy, their robustness declines once exposed to new subjects, sessions, tasks, or paradigms. Cross-task generalization is only partially successful when cognitive mechanisms overlap, as in Stroop and Mirror, but fails for tasks that rely on fundamentally different processes such as arithmetic. Notably, although individual tasks in Dataset B yielded lower scores in isolation, their combination produced the strongest overall performance , suggesting that aggregating cognitively diverse paradigms enhances the salience of attention-related neural markers. Most critically, cross-dataset/cross-paradigm transfer remains an unsolved problem, as paradigm-specific variability introduces distributional shifts that substantially limit the effectiveness of conventional classifiers. evaluation, but limited the volume and diversity of data available for model fitting. The analysis also used the 12 shared channels, a fixed parsimonious preparation pipeline, and predeclared model recipes without hyperparameter or architecture selection to provide comparable pre-adaptation benchmarks rather than to maximize performance in each scenario. The reported figures consequently do not represent the maximum attainable performance for an individual scenario.

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Table 1. Unified binary attention label mapping across datasets.

Attention State	Conditions
High Attention	Focused, Stroop, Arithmetic, Mirror
Low Attention	Unfocused, Rest

Table 2. Analytical dataset compositions.

Dataset configuration	Retained conditions	Participants	Recording units	Windows, n (low / high)	Class distribution, % (low / high)
Full Dataset A	Focused, Unfocused	5	24	3,192 / 3,192	50.0 / 50.0
Full Dataset B	Rest, Stroop, Arithmetic, Mirror	40	120	600 / 1,800	25.0 / 75.0
B: Stroop	Rest, Stroop	40	120	600 / 600	50.0 / 50.0
B: Arithmetic	Rest, Arithmetic	40	120	600 / 600	50.0 / 50.0
B: Mirror	Rest, Mirror	40	120	600 / 600	50.0 / 50.0

Table 3. Performance metrics across experimental scenarios for Logistic Regression.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.64	0.63	0.30	0.73	0.62/0.66
	Stratified K-fold(Full B)	0.68	0.64	0.34	0.74	0.65/0.71
	Stratified K-fold(B: Stroop)	0.69	0.68	0.40	0.74	0.67/0.71
	Stratified K-fold(B: Arithmetic)	0.66	0.65	0.35	0.74	0.66/0.67
	Stratified K-fold(B: Mirror)	0.69	0.67	0.39	0.75	0.66/0.72
Cross-subject	Leave-one-subject-out(Full A)	0.63	0.61	0.29	0.73	0.63/0.63
	Leave-one-subject-out(Full B)	0.66	0.61	0.31	0.74	0.63/0.69
Cross-task (Dataset B)	Full B→Stroop	0.66	0.64	0.34	0.73	0.64/0.68
	Full B→Arithmetic	0.64	0.62	0.31	0.72	0.64/0.65
	Full B→Mirror	0.67	0.65	0.36	0.77	0.63/0.71
	Stroop→Full B	0.63	0.58	0.25	0.67	0.63/0.63
	Stroop→Arithmetic	0.58	0.55	0.16	0.60	0.65/0.50
	Stroop→Mirror	0.63	0.60	0.27	0.68	0.64/0.62
	Arithmetic→Full B	0.62	0.55	0.23	0.68	0.66/0.58
	Arithmetic→Stroop	0.56	0.53	0.14	0.61	0.67/0.46
	Arithmetic→Mirror	0.66	0.64	0.35	0.73	0.67/0.66
	Mirror→Full B	0.64	0.57	0.26	0.70	0.68/0.60
	Mirror→Stroop	0.60	0.57	0.21	0.65	0.66/0.54
	Mirror→Arithmetic	0.62	0.59	0.27	0.70	0.63/0.62
	Full A→Full B	0.52	0.37	0.03	0.54	0.78/0.26
	Full B→Full A	0.55	0.53	0.12	0.58	0.35/0.76
	Full A→Stroop	0.49	0.42	-0.03	0.51	0.71/0.27
Cross-dataset	Stroop→Full A	0.54	0.53	0.08	0.56	0.45/0.63
	Full A→Arithmetic	0.53	0.48	0.08	0.58	0.68/0.38
	Arithmetic→Full A	0.52	0.52	0.04	0.54	0.43/0.61
	Full A→Mirror	0.53	0.47	0.06	0.55	0.72/0.34
	Mirror→Full A	0.52	0.48	0.06	0.54	0.26/0.79

Fig 1. Zero-shot transfer performance across source and target compositions. Rows denote a decoder and the composition it was trained on, and columns denote the composition it was tested on. The gray diagonal is not a transfer but the baseline validation scenario, where a decoder was trained and tested on the same composition. The two right-hand columns show mean differences from the relevant target baselines. Negative values indicate lower performance, and dashes mark contrasts that are not applicable.

Fig 2. Baseline and cross-subject performance across decoders.

Fig 3. Participant-level balanced accuracy across experimental scenarios for the five decoders.

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Supporting information

S1 Table. **Performance metrics across experimental scenarios for XGBoost.**

S2 Table. **Performance metrics across experimental scenarios for EEGNet.**

S3 Table. **Performance metrics across experimental scenarios for ShallowFBCSPNet.**

S4 Table. **Performance metrics across experimental scenarios for EEGConformer.**